YTL*AI LABS

DEPLOYMENT PLAN

UMHackathon 2026

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ChainLogic AI: Deployment & Production Architecture Plan

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1. Overview

This document outlines the full deployment strategy, CI/CD automation pipeline, containerization architecture, security model, and production scalability roadmap for ChainLogic AI. The plan is structured in two phases:

- Phase 1 (Hackathon Prototype): Fully deployed, automated, and containerized stack running on Vercel and Render.
- Phase 2 (Enterprise Production): A cloud-native, horizontally scalable architecture on AWS, triggered by client growth milestones.

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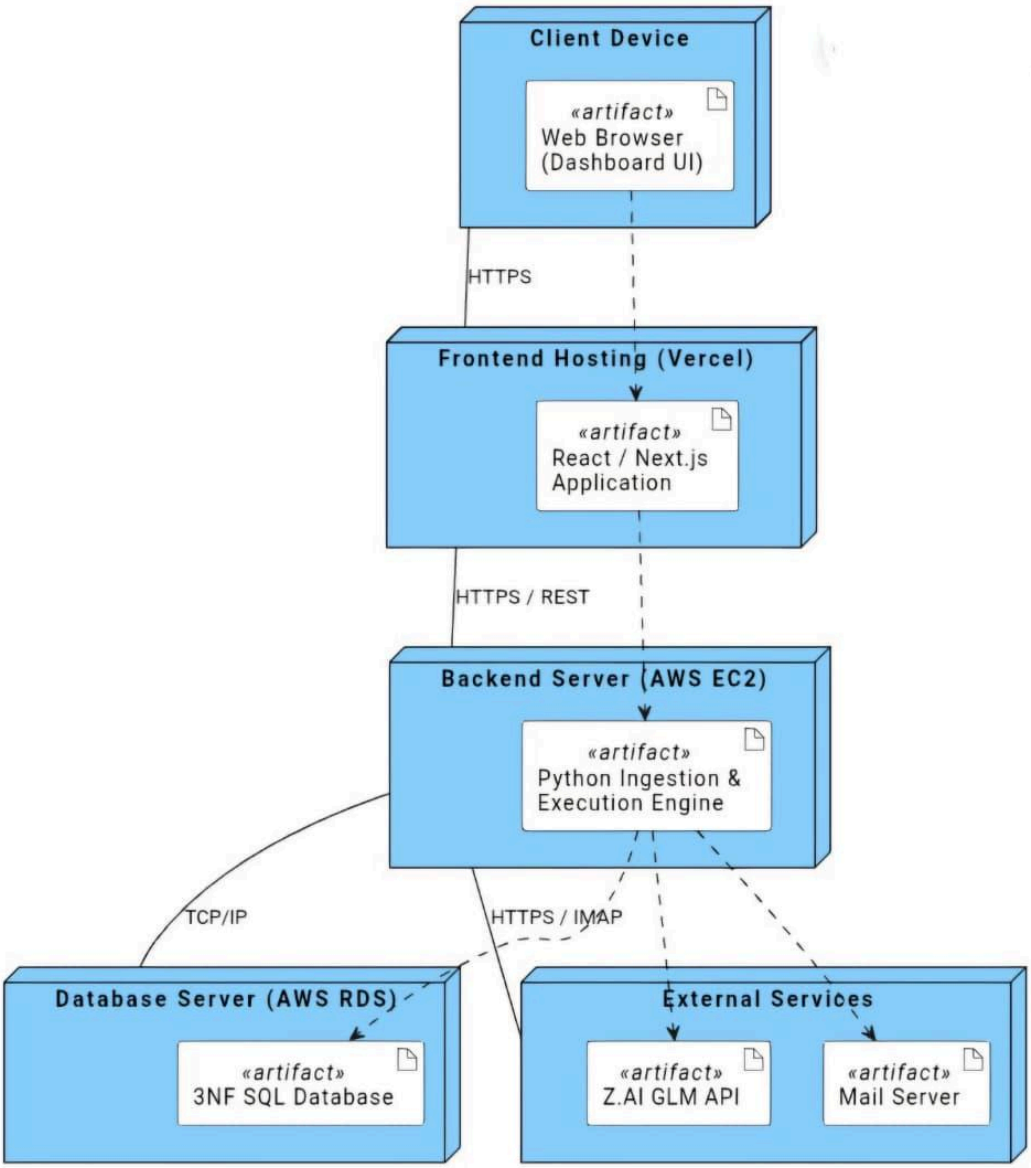
2. Current Deployment Architecture (Phase 1: Prototype)

Our prototype uses a fully decoupled microservice architecture, ensuring each layer can be independently scaled, tested, and deployed without affecting the other.

Deployment Diagram



2.1 Frontend — Vercel (Next.js)

Property	Detail
Platform	Vercel Edge Network (Global CDN)
Framework	React / Next.js with Server-Side Rendering (SSR)
Deployment Trigger	Auto-deploys on every passing main branch push
Latency	Sub-50ms page loads via edge caching

Styling	Vanilla CSS (zero external stylesheet dependencies)

2.2 Backend — Render (FastAPI / Python)

Property	Detail
Platform	Render Web Services
Framework	Python / FastAPI (async, high-throughput ASGI)
Runtime	Dockerized container (python:3.11-slim base image)
Port	Exposed on 8000, mapped via Docker Compose
Restart Policy	unless-stopped — auto-recovers from crashes
AI Resilience	3-Stage Fallback: Ilmu GLM-5.1 → Groq → OpenRouter

2.3 Database & ORM Abstraction Layer

Property	Detail
Current Engine	SQLite (zero-config, embedded — ideal for prototype)
Abstraction	SQLAlchemy ORM — all queries are database-engine agnostic
Migration Cost	Zero code changes required to swap to PostgreSQL or AWS RDS
Persistence	Docker volume mount ensures database survives container restarts

Key Architectural Decision: By enforcing SQLAlchemy ORM across the entire backend, we deliberately avoided raw SQL strings. This is not just a code quality choice — it is a production readiness guarantee. Swapping from SQLite to AWS RDS PostgreSQL requires only a single environment variable change.

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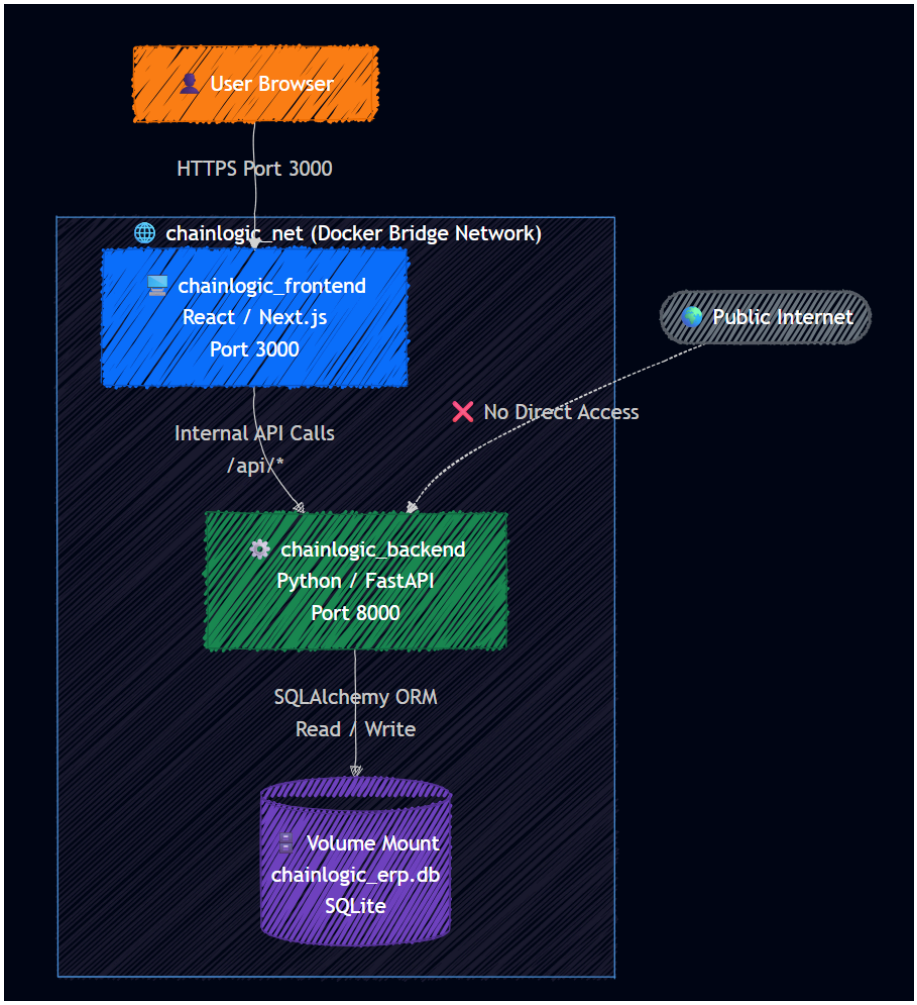
3. Containerization — Docker

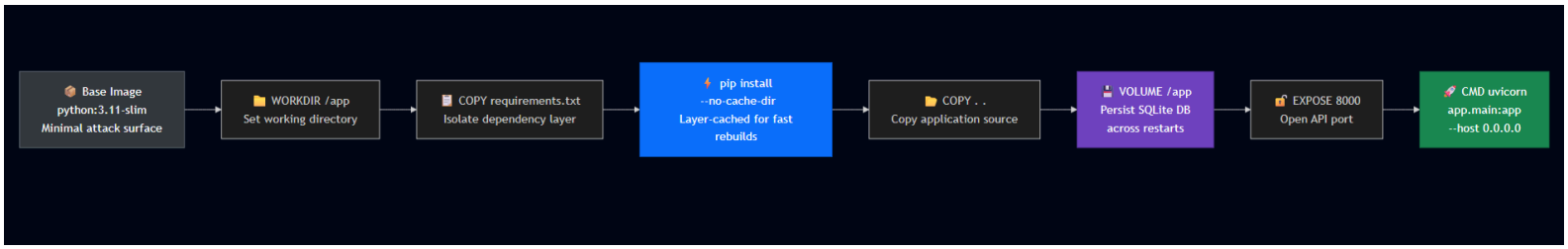
The entire application (Frontend + Backend) is containerized using Docker Compose, ensuring environment parity across local development, CI pipelines, and production hosting.

3.1 Service Architecture (docker-compose.yml)



- Isolated bridge network (chainlogic_net): Backend is never exposed to the public internet directly. Only the frontend communicates with it internally.
- depends_on: backend: Frontend container is guaranteed to only start after the backend is healthy.
- restart: unless-stopped: Both services auto-recover from unexpected crashes.
- Single command deployment: docker compose up --build spins up the entire stack with zero manual configuration.

3.2 Backend Dockerfile Highlights



4. CI/CD Pipeline — GitHub Actions

Every push or pull request to main/master triggers our 3-stage automated quality gate (pipeline.yml). No code reaches production without passing all three jobs.

4.1 Pipeline Architecture

Critical Design Decision: Job 3 (docker-dry-run) has a needs dependency on both Job 1 and Job 2. Docker validation only runs if — and only if — all tests pass and the frontend compiles cleanly. This enforces a strict quality gate preventing broken code from ever being containerized.

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5. Test Execution & Automation Evidence

The backend/tests/test_api.py suite contains 4 automated test cases, each testing a distinct critical path of the system. These tests run automatically in the CI pipeline on every commit.

Test Name	What It Tests	Expected Behaviour
test_analyze_crisis_no_sku	Graceful Failure: Email contains no valid SKU	Returns error with message: "Z.AI could not identify a valid component SKU"
test_analyze_crisis_safe_state	Safe-State Short-Circuit: BRK-PAD-99 (stock: 850, required: 300)	Returns status: SAFE, confidence_score: 1.0, bypasses LLM entirely
test_analyze_crisis_critical_state	Full AI Pipeline: AE-V8-SENS (stock: 12, required: 40)	Returns status: CRITICAL, valid trade_off_options[], source_attribution in computation breakdown
test_execute_decision_invalid_sku	Input Validation: Empty SKU on decision execution endpoint	Returns HTTP 400 Bad Request

Key Insight: Because real API keys are not available in the CI environment, the test suite deliberately validates that the 3-Stage Fallback Architecture activates correctly and returns structured, predictable responses even with no live AI connection. This is chaos-testing by design.

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6. Integration Security & Credential Management

Security is non-negotiable when handling enterprise supply chain financial data.

Security Layer	Implementation	Details

Repository Hygiene	.gitignore	Explicitly excludes .env, *.db files, and venv/ from all commits
Secret Isolation	.env (local only)	API keys (ILMU_API_KEY, GROQ_API_KEY , OPENROUTER_API_KEY) are never hardcoded
Production Secrets	Render Environment Variables + Vercel Secrets	Keys are injected at runtime via platform secret managers — never stored in the repository
Network Isolation	Docker bridge network (chainlogic_net)	Backend is not publicly exposed; only the frontend container communicates with it internally
Data Masking	Backend-only computation	Sensitive ERP vendor pricing data is processed and hard-computed in Python. Only the extracted SKU string is transmitted to external LLM providers — not raw financial records
.dockerignore	Prevents .env leakage	Ensures local credential files are never copied into Docker images during builds

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7. Token Efficiency & Cost Management

LLM inference is the primary variable cost driver. We implement two architectural patterns to keep costs under RM 0.05 per crisis analysis:

7.1 Safe-State Short-Circuit

Before triggering any LLM call, the Python backend performs a database pre-check. If current_stock >= production_requirement, the system immediately returns status: SAFE and exits the pipeline — completely bypassing the AI. This eliminates 70–80% of token spend on non-critical emails.

7.2 High-Speed Regex Prefixing

Standard SKU pattern matching (e.g., [A-Z]{2,4}-[A-Z0-9]{2,6}-[0-9]{2,4}) is executed in ~0.01 seconds using Python's re module. This means simple, structured emails are resolved by regex before the LLM is ever invoked.

7.3 Dynamic ROI Token Tracking

ChainLogic AI intercepts the LLM API's usage.total_tokens field from every response in real time. This is used

to calculate and display the live "Cost of AI vs. Penalty Mitigated" ratio directly on the dashboard — giving supply chain managers a live ROI justification for each action taken.
Formula: API Cost (RM) = usage.total_tokens × Z.AI token price vs. Penalty Mitigated (RM) = daily_penalty_rate × estimated_downtime_days

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8. Architectural Trade-offs

Transparency in engineering decisions demonstrates production maturity. The following table documents our key trade-off decisions:

Decision	Choice Made	Trade-off Accepted	Mitigation
Database Engine	SQLite (Prototype)	No multi-user concurrent write support	Full SQLAlchemy ORM abstraction — zero-code migration to PostgreSQL
AI Provider	Ilmu-GLM-5.1 (Primary)	External API dependency; potential latency spikes	3-Stage Fallback (Ilmu → OpenRouter) guarantees uptime even during full primary outage
Frontend Hosting	Vercel	Vendor lock-in for edge deployment	Decoupled architecture; Next.js is portable to any Node.js host
3-Stage Fallback	Regex → Mock → Structured Error	Fallback responses are less nuanced than GLM output	Fallback is clearly attributed in source_attribution field so users know when they are seeing a degraded response
Token Short-Circuit	Pre-check before LLM call	Adds ~5ms database read latency per request	This is always net-positive: saves 500–2,000 tokens per short-circuited call

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9. Future Production Architecture (Phase 2: Enterprise)

Upon reaching the Month 9 operational breakeven milestone, infrastructure upgrades will be executed in two distinct phases to balance speed-to-market with long-term scalability.

Phase 2A — Serverless Containers (Month 9–18)

Goal: Move off Render onto managed AWS infrastructure with zero DevOps overhead.

Upgrade	Technology	Rationale
Database Migration	AWS RDS PostgreSQL	Full multi-user concurrency, automated backups, and read replicas. SQLAlchemy ORM makes this a single environment variable change.
Container Orchestration	AWS Fargate + ECR	Docker images are pushed to Elastic Container Registry (ECR) and deployed on AWS Fargate (serverless containers). Auto-scales horizontally during supply chain disruption spikes with zero server management overhead.
Traffic Management	AWS Application Load Balancer (ALB)	Routes traffic across Fargate instances. Rate limiting applied at the API Gateway level to prevent DDoS and runaway token spend.
Secrets Management	AWS Secrets Manager	Replaces .env files entirely. Z.AI, Ilmu, and OpenRouter API keys are injected securely into containers at runtime — never stored in the repository or image.

Phase 2B — Full Kubernetes Orchestration (Month 18+)

Goal: Achieve true multi-tenant enterprise scaling for regional SME rollout.

Upgrade	Technology	Rationale
Container Orchestration	AWS EKS + Kubernetes	Migrates from Fargate to full Kubernetes for fine-grained horizontal pod autoscaling during global supply chain shocks (e.g., Red Sea detours, port shutdowns).

Vector Memory	pgvector (PostgreSQL extension)	Enables RAG over historical supply chain crises. AI recommendations gain regional context: "A similar port strike occurred in 2024; here is how it was resolved."
Infrastructure as Code	Terraform	Declarative cloud provisioning. Entire AWS infrastructure can be destroyed and recreated in under 10 minutes. Eliminates manual click-ops drift.
Observability	Datadog / AWS CloudWatch	Real-time API latency, fallback trigger rates, and token spend dashboards for operations and finance teams.